Name: ____________________________

Exit quiz

Problem solving with non-linear relationships

1 This table shows the value of a car two years after it was made. If the values form a geometric sequence, what is the common multiplier between terms? (You may use your calculator.) Tick **1** correct answer

Year	Value of car (£)
0	20 000
1	16 000
2	12 800
3	

☐ 0.2

☐ 0.4

☐ 0.6

☐ 0.8

☐ 1.25

2 This table shows the value of a car 2 years after it was made. If the values continue to follow the same geometric sequence, what is the car's value after 3 years? (You may use your calculator.) £ ____________________ Fill in the blank

Year	Value of car (£)
0	20 000
1	16 000
2	12 800
3	

3 This table shows the value of a car two years after it was made. Why might this sequence not continue as the same geometric sequence forever? Tick **2** correct answers

Year	Value of car (£)
0	20 000
1	16 000
2	12 800
3	

- ☐ If the geometric sequence continues the value will drop below zero pounds.
- ☐ Sequences cannot go on forever, they will eventually stop.
- ☐ The value of the car may stop decreasing, it could reach minimum value.
- ☐ The value may decrease sharply in the first 2 years, then the pattern may change
- ☐ The car loses the most value in the first year, not the same amount each year.

4 Which two sequences can be added to make the first 5 terms of a linear sequence? Tick **2** correct answers

- ☐ 3, 7, 11, 15, 19, ...
- ☐ 5, 10, 20, 40, 80, ...
- ☐ 6, 7, 9, 12, 16, ...
- ☐ 14, 11, 8, 5, 2, ..
- ☐ 16, 24, 36, 54, 81, ...

5 Order these terms so they form part of a Collatz sequence: "If the term is even, the next term is half the previous term. If the term is odd, the next term is 3 times the previous term add 1". Use numbers to show the correct order

	100
	25
	33
	76
	38
	50

6 Which two sequences can be added to make the first 5 terms of a linear sequence? Tick **2** correct answers

- [ ] 1, 3, 8, 16, 27, …
- [ ] 8, 11, 13, 14, 14, …
- [ ] 10, 19, 25, 28, 28, …
- [ ] 15, 16, 19, 24, 31, …
- [ ] 20, 17, 14, 11, 8, …